

SleepyFlora: Supporting Sleep Sharing and Augmentation over a Distance for Social Bonding across Time Zones

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ABSTRACT

Sleep plays a significant role in our health and well-being. We can see an increasing amount of work in HCI exploring the design for better sleep. However, most of these works only focus on sleep tracking and reminding. The social perspective of sleep is rarely explored. We believe that sleep is tightly associated with social connection and intimacy. This work explores how sleep sharing can facilitate social bonding over a distance. We present SleepyFlora, an interactive system that enables a user to know the other user's real-time sleep stages via a flower-shaped installation and augment their sleep by sending soothing aromas and playing music remotely. With SleepyFlora, we hope to further the understanding of interaction design around sleep by considering its social perspective. Moreover, we hope our work can provide a novel approach, i.e., enabling sleep sharing, for facilitating social bonding over a distance.

CCS CONCEPTS

• **Human-centered computing** → **Collaborative interaction**.

KEYWORDS

Sleep, Sleep Stages, Social Bonding, Interaction Design, Computational Sleep

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1 INTRODUCTION

Recently, HCI research has increasingly focused on designing interactive technologies to enhance sleep, given its crucial role in maintaining physical and mental health [19]. Adequate sleep positively impacts cognitive functions such as learning and memory [28], while sleep deprivation is associated with various health issues and cognitive impairments, including increased risk of obesity, heart diseases, depression, and problems with attention, concentration, and decision-making [2, 20]. Consequently, HCI researchers are investigating how to develop technologies that effectively promote healthy sleep.

Current research predominantly focuses on sleep tracking support [4] and nighttime sleep intervention using interactive technologies, e.g., designing external stimulations to intervene one's sleep and dream [9]. Nevertheless, the social dimension of sleep has been largely overlooked. Despite seemingly hindering real-time communication, sleep is closely linked to social connection and intimacy. This connection is exemplified by synchronous sleep patterns among cohabiting couples, nightly greetings among loved ones, and bedtime stories for children. These examples suggest potential opportunities for leveraging the social perspective of sleep, to enhance social connections. Notably, these examples pertain to synchronous interactions within the same time zone. In this study, we investigate the design of sleep technologies to promote social bonding across different time zones.

We introduce SleepyFlora, a sleep technology facilitating remote sleep sharing. The system comprises an EEG headset and two desktop installations in the users' homes. The headset collects brain signals, with a machine-learning algorithm classifying real-time sleep stages from the EEG data. SleepyFlora wirelessly transmits this information to a flower-shaped installation in the awake user's home, using colored light to indicate sleep stages. The awake user can remotely control the second installation in the sleeping user's home, releasing soothing aromas or playing music based on real-time sleep status for improved sleep and wake-up reminders.

Our work makes the following contributions.

- Introducing SleepyFlora, a sleep sharing and augmentation system promoting social bonding for users across time zones.

- Leveraging the social perspective of sleep to expand the current design of social interactions over a long distance.
- Calling for future work to consider the social affordances of sleep in interaction design.

We aim for SleepyFlora to strengthen social bonds among users in different time zones and provide insights into sleep technology's social aspects. We hope to motivate researchers to explore sleep's social importance for enhancing sleep and dream experiences, as well as inspire those interested in remote presence and long-distance connections to consider sleep in their designs.

2 BACKGROUND: SLEEP CYCLE AND SLEEP STAGES

In our work, we designed SleepyFlora to measure the user's real-time sleep stages. This section gives a brief introduction of sleep cycles and sleep stages.

A sleep cycle, characterized by an oscillation between slow-wave and REM sleep phases [30], typically lasts 70-110 minutes, averaging 90 minutes. Ideally, an individual undergoes four to six sleep cycles per night, comprising four stages: three non-rapid eye movement (NREM) and one rapid eye movement (REM) sleep, each exhibiting distinct brain activity patterns.

Stage one (N1) is a brief, 1-5 minute period of drowsiness or initial sleep with reduced brain activity, partial body relaxation, and easy arousal. Progressing to stage two (N2), light sleep ensues, with ceased eye movement, slower brain activity, and further relaxation, as well as decreased body temperature and heart rate. N2 sleep lasts 10-25 minutes during the initial cycle, gradually increasing in subsequent cycles, occupying about half of the total sleep time. Stage three (N3) is the deepest sleep stage, marked by delta wave brain activity, heightened relaxation, and further reduced heart and breathing rates, rendering arousal difficult. N3 sleep duration ranges from 20-40 minutes in early cycles and shortens in later cycles. Stage four, REM sleep, features brain activity resembling wakefulness, typically lasting 10 minutes, with the final REM stage extending up to an hour.

Monitoring sleep cycles is crucial for evaluating sleep quality and diagnosing sleep disorders [32]. Researchers exploring sleep stage classification methods, employing biosignals such as electroencephalogram (EEG), electrooculogram (EOG), electromyogram (EMG), and electrocardiogram (ECG) signals to identify sleep patterns [5]. Among these biosignals, EEG proves most reliable for sleep stage identification [18], and we use it in this work to determine users' real-time sleep stages.

3 RELATED WORK

Our work builds on prior works around technology-mediated social connections and sleep in HCI.

3.1 Technology Mediation for Social Connection over a Distance

Designing technologies for long-distance social connections has been a focus of HCI research for decades. Some studies enhance social connections by expanding interactions temporally and spatially.

Lottridge et al. [17] explored sharing empty moments without explicit messages, such as asymmetrical music and background sound sharing for remote couples. Yang and Neustaetter [33] investigated telepresence robots for communication, with the robot's physicality and mobility promoting a sense of co-presence.

Some studies focus on enabling users to experience physical intimacy remotely, such as hugs, touch, and kisses. Mueller et al. [22] introduced an air-inflatable vest, Hug Over a Distance, simulating a hug sensation. Samani et al. [25] developed Kissenger, a device for transmitting kisses between remote individuals. Kontaris et al. [13] presented Feelybean, allowing remote partners to experience each other's touch. These works highlight the potential for technology to mediate physical sensations, effectively enhancing social connections, primarily targeting remote couples due to the intimacy involved.

Biosignals and emotion sharing is another common approach for improving connectedness [31]. As biosignals are intimate and social-emotional, sharing them can foster deep emotional attachment [7]. Janssen et al. [11] studied how sharing heartbeat sounds enhances intimacy and closeness. Liu et al. developed a mobile app enabling real-time heart rate sharing via text messaging. Hassib et al. [10] introduced EmotionActuator, a system transmitting emotional states, actuating the recipient's muscle through electrical muscle stimulation (EMS) for empathetic understanding of the sender's emotions. These studies highlight the potential of biosignals and emotions for facilitating intimate and playful social interactions.

In addition to the aforementioned strategies, some studies focus on sharing daily activities to improve social connectedness. Chung et al. [6] designed paired cups for users to share drinking times across distances. Tibau et al. [29] introduced FamilySong, a prototype enabling synchronized music sharing among internationally distributed family members, fostering togetherness. Inspired by these studies, we similarly enable users to share daily activities, specifically sleep, with close individuals.

3.2 Sleep in HCI

Early HCI works on sleep primarily focus on sleep tracking. Min et al. [21] introduced a mobile app utilizing sensor data for detecting sleep states and daily sleep quality. Choe et al. developed SleepTight, an Android widget displaying sleep data on lock and home screens to encourage reflection on sleep behaviors. Koushik et al. [14] proposed a real-time sleep stage monitoring mobile application based on users' EEG signals.

Building on sleep tracking technologies, HCI researchers explore sleep monitoring by raising user awareness of sleep patterns and intervening in sleep behaviors [4]. Ploderer et al. [24] introduced SleepBeta, a web-based app for teenagers' self-reflection on sleep habits. Lee et al. [16] developed SleepGuru, an actionable sleep planning system offering personalized sleep management plans. Pai et al. [23] presented NapWell, a virtual reality sleep assistant to reduce sleep onset latency.

Recent works explore cognitive perspectives, designing technology to augment one's sleep quality and cognitive performance [27]. Amores et al. [1] introduced Ezzenze, a smartphone-based olfactometer that can release scents based on sleep stages. Horowitz

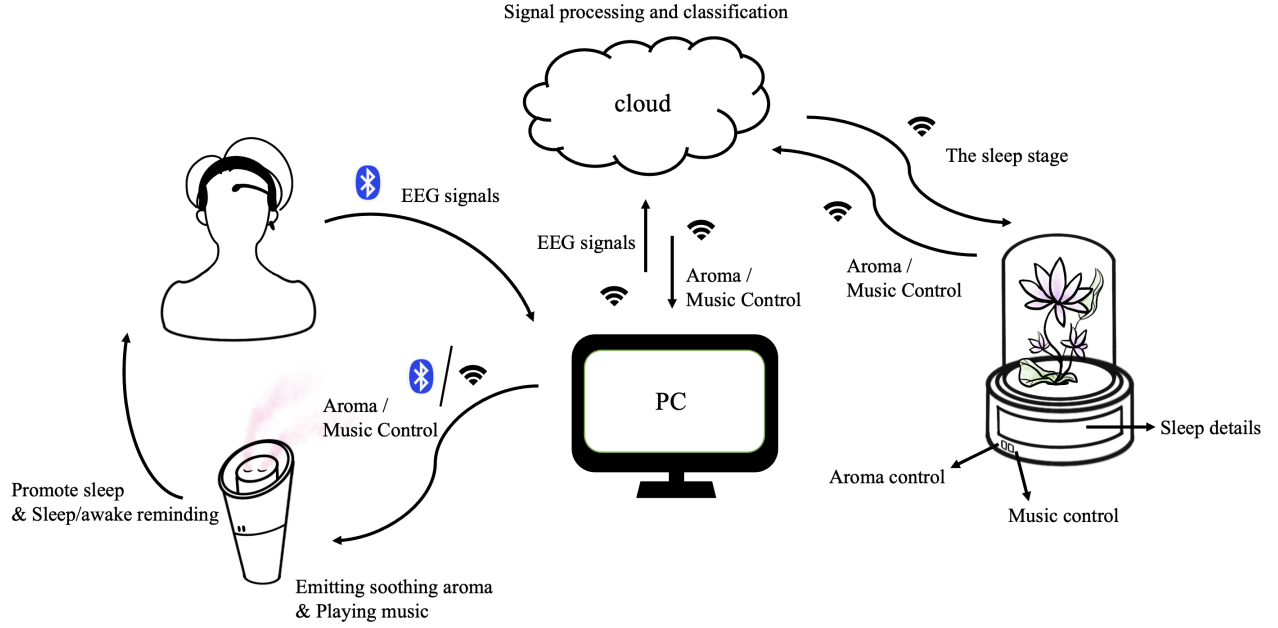


Figure 1: The interaction diagram of SleepyFlora

et al. [9] examined the design space of Hypnagogia, a consciousness state stimulating creativity. Carr et al. [3] discussed sensory stimulation technologies' potential in dream engineering and sleep manipulation.

Few studies in the field have investigated sleep in a social context. Gooch and Watts [8] presented SleepyWhispers, a prototype enabling good-night audio message sharing. Kučera et al. [15] introduced Bedtime Window for long-distance couples to share sleep photos and collaborate on drawings. Schmidt [26] and Kim et al. [12] both presented alarm clocks incorporating social network information to inform call appropriateness or alarm settings.

In summary, most existing works view sleep as an obstacle to social interactions, focusing on design spaces before and after sleep or sleep state sharing for interaction initiation. Our work explores interaction design during sleep, potentially providing novel opportunities to improve social bonding over a distance, particularly for individuals with different sleep schedules due to time differences.

4 SLEEPYFLORA

SleepyFlora is a two-user household system allowing users to know and augment close friends or family members' sleep states remotely. We refer to the two users as 'sleeper' and 'awaker'. The awaker can know the sleeper's real-time sleep stage using a lotus-shaped installation with color-changing LEDs. The installation also features a control module for the awaker to remotely operate an aroma diffuser in the sleeper's home, emitting soothing aromas for better sleep. The diffuser also includes a speaker, enabling the awaker to play sleep or wake-up reminder music.

4.1 System Design and Implementation

SleepyFlora consists of three parts: a *sleepy end* with devices in the sleeper's house, a cloud server, and a *flora end* with devices in the awaker's house (see Figure 1). The sleepy end includes an EEG headset and an aroma diffuser with a speaker. The headset collects the sleeper's brain signals for sleep stage classification, while the diffuser is remotely controlled by the awaker for aroma emission and music playback. The cloud server processes data and transfers signals between both ends. The flora end features a lotus-shaped installation with a control module, displaying soft light indicating the sleeper's real-time sleep stage. The awaker uses the control module to remotely operate the aroma diffuser at the sleepy end.

In this section, we elaborate on the design and technical details of our system implementation.

4.1.1 Sleepy End.

We selected the Mindwave Mobile2 EEG headset by NeuroSky for EEG signal acquisition (Figure 2a). This low-cost wearable device features flexible rubber sensor arms with electrodes, a forehead sensor tip, a T-shaped headband, and an ear clip. The headset enables wireless real-time signal transmission via Bluetooth, allowing us to send EEG data to a PC and subsequently to a cloud server.

Additionally, we developed a diffuser incorporating a speaker for emitting aromas and playing music (Figure 2b). The system uses an Arduino-compatible water atomization driver board with cotton soaked in lavender essential oil and an ultrasonic transducer plate. The setup is connected to an Arduino UNO with an HC-05 Bluetooth Module. When prompted by the cloud server, the PC sends a signal to the Arduino to activate the diffuser and play music through an external speaker.

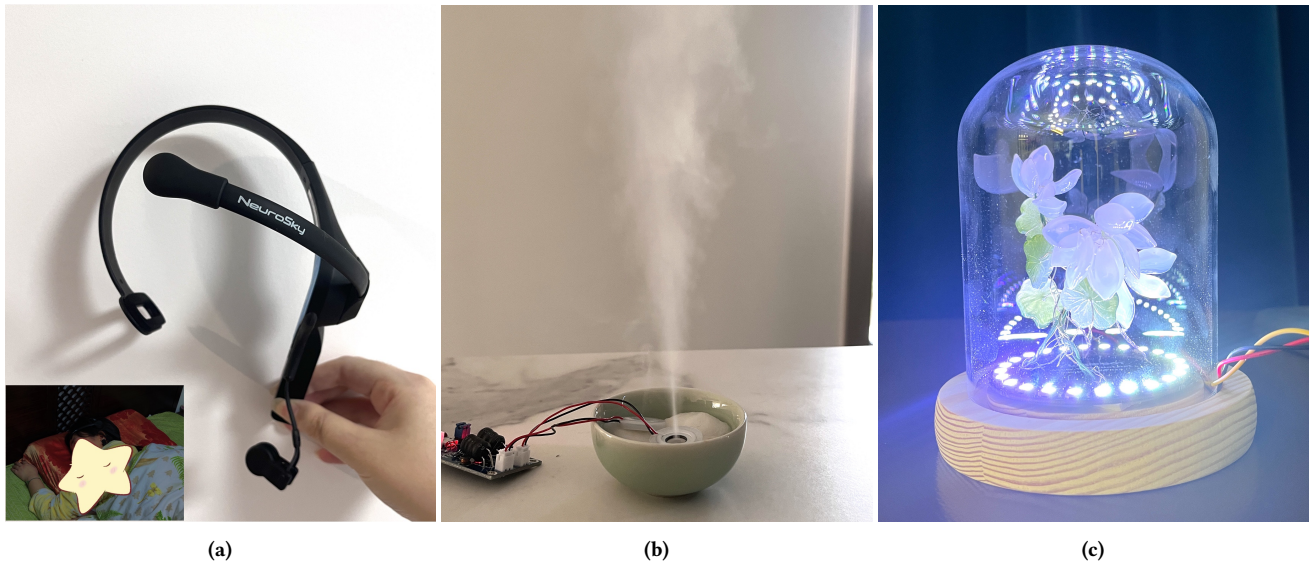


Figure 2: (a) The Mindwave Mobile2 EEG headset. (b) The prototype of the aroma diffuser. (c) The lotus-shaped installation deployed at the awaker's home.

4.1.2 Flora End.

The Flora end, situated at the awaker's home, includes a desktop installation for visualizing the sleeper's real-time sleep stage and a control module for aroma emission or music playback (Figure 2c). The installation features a wood base, glass dome, and artificial lotus flowers and leaves, symbolizing peace, purity, and beauty in the regional culture.

To realize the installation, a WS2812 module with 24 multicolor LEDs connects to an Arduino UNO with an ESP8266 WiFi Module, receiving sleep stage data from the cloud server. LEDs change color based on the sleep stage: off when awake, and transitioning through red, green, blue, and white for different stages. A mini-screen displays sleep data summaries after the sleeper awakens. The control module, consisting of two touch sensors connected to the Arduino, activates smell emission and audio playback, sending signals to the cloud server upon touch detection.

4.1.3 Cloud Server.

The cloud server preprocesses raw EEG signals and classifies sleep stages using our designed MultiConvSleepNet model, based on single-channel EEG data. The model comprises a representation learning module for automatic feature detection, a sequence learning module for timing information, and a softmax classifier for final sleep stage classification. Further details are available in our open-source Github repository¹.

We trained MultiConvSleepNet on the Sleep-EDF-39 dataset, a subset of Sleep-EDF, containing 42,308 30-second sleep epochs from 20 healthy subjects. We applied 20-fold cross-validation to evaluate our model and chose the weighted cross-entropy loss function to alleviate sample imbalances among sleep stages. Training occurred over 100 epochs on an NVIDIA Tesla K80, achieving an overall accuracy of 86.6%.

5 PROGRESS AND FUTURE WORK

At this stage, the cloud server module has been completed. With the sleepy end, we have successfully enabled real-time collection of EEG signals, their transmission to the cloud server, and the circuit design of the aroma diffuser. We will assess the comfort of the EEG headset during sleep, considering alternative headbands if participants report any discomfort. Additionally, we will design and 3D print a case for the aroma diffuser to finalize the product. With the flora end, we have completed the design of the flora-shaped LED installation and the circuitry for the control module. Our next steps involve integrating the control module into the installation and finalizing the overall product design.

Once the system development is complete, we will conduct a user study to understand the value of SleepyFlora for individuals with close relationships residing in different time zones. A pilot study with two pairs of participants will provide feedback for system iteration. Subsequently, an empirical study with approximately six pairs will assess the user experience over two weeks. We will use a combination of survey and semi-structured interviews to gather insights on participants' engagement with the technology and its impact on their social connectedness. We also acknowledge potential privacy concerns associated with a social sleep system. SleepyFlora is designed for individuals with close relationships who are comfortable sharing private sleep information. Nonetheless, we will collect data regarding participants' privacy concerns and their willingness to share sleep information based on the extent of their close relationship with others. Finally, through data analysis, we will articulate SleepyFlora's user experience and provide guidance on future interaction design for social sleep.

¹<https://github.com/czhczh111/sleep-stage-scoring-multiConvSleepNet>

6 CONCLUSION

Sleep plays a crucial role in health and wellbeing, attracting HCI research attention. However, the social aspect remains underexplored. We present SleepyFlora, an interactive system enabling sleep stage sharing between users over distance, targeting those in different time zones. SleepyFlora allows users' close friends or family members to monitor their real-time sleep stages and augment their sleep through music or soothing smells. With this work, we aim to enhance social connections, inspire researchers to consider the social perspective of sleep, and explore its potential as a design material for engaging social experiences.

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